

1- a)

$$= \bar{x}\bar{y} + xz$$

b)

$$= \bar{x}z + y$$

2- a)

$$= xy + \bar{x}z$$

b)

$$= \bar{x} + yz$$

c)

$$= \bar{c} + \bar{a}b$$

3- a)

$$= \bar{a}\bar{b} + cdb$$

b)

$$= w\bar{x} + \bar{w}xy$$

c)

$$= cd + abd + abc$$

5- a)

$$= z + \bar{x}y$$

b)

$$= \bar{b}\bar{c} + \bar{b}d + a\bar{b}c$$

c)

$$= \bar{b}d + cd + ac + \bar{a}bd$$

d)

$$= \bar{x}y + wz + yz + wy$$

4- a)

$$= \bar{y}z\bar{x} + \bar{y}z\bar{w} + wx\bar{y} + yz\bar{x}$$

b)

$$= \bar{a}\bar{b}\bar{d} + \bar{a}\bar{c} + \bar{a}bd + acd$$

c)

$$= \bar{x}y + wx$$

d)

$$= bd + \bar{a}\bar{d} + \bar{b}\bar{d}$$

6- a)

AB \ CD	00	01	11	10
00			1	
01		1	1	1
11				
10				

$$= \Sigma(3, 5, 6, 7)$$

AB \ CD	00	01	11	10
00			1	1
01			1	
11	1	1		1
10			1	

$$= \Sigma(1, 3, 5, 9, 12, 13, 14)$$

AB \ CD	00	01	11	10
00	1			1
01			1	1
11			1	1
10	1			1

$$= \Sigma(0, 2, 6, 7, 8, 10, 14, 15)$$

8- a)

AB \ CD	00	01	11	10
00				
01	1	1		
11				
10				

$E=0$

$$= \bar{A}\bar{C}\bar{E} + \bar{C}DE + \bar{A}\bar{B}\bar{C}$$

AB \ CD	00	01	11	10
00	1			
01		1		
11			1	
10			1	

$E=1$

AB \ CD	00	01	11	10
00	1	1	1	1
01	1	1	1	1
11			1	
10			1	

$E=0$

$$= \bar{A}\bar{B}\bar{D} + CD + \bar{A}B\bar{E}$$

AB \ CD	00	01	11	10
00	1	1	1	1
01			1	1
11			1	
10			1	

$E=1$

c)

AB \ CD	00	01	11	10
00	1			
01		1	1	
11			1	
10	1			1

$E=0$

$$= \bar{B}\bar{D}\bar{E} + B\bar{D}\bar{E} + \bar{B}C\bar{E} + \bar{A}\bar{B}C\bar{D} + \bar{B}C\bar{D}$$

AB \ CD	00	01	11	10
00	1			
01				
11				
10				

$E=1$